Numerical Simulation Methods And Their Applications

Monte Carlo Methods and Applications in Particle-based Systems

Ridhwaan Amin
32005096

Department of Mathematics and Statistics
University of Reading

February 21, 2026

Plain language summary

This project concerns working with a computer simulation of a particle-based system, particularly using numerical techniques known as Monte Carlo simulations.

It starts with an introduction to the history of these methods, touching upon the infamous Manhattan Project, then focusing on pseudo-random number generators, sampling methodologies, and their applications in statistical mechanics. The project also explores the Metropolis method and its application to a Lennard-Jones liquid. Finally, it concludes with findings of a simulation created from individually developed C++ code.

Declaration

I certify that this is my own work and that use of material from other sources has been properly and fully acknowledged in the text. I have read the University's definition of plagiarism and the department's advice on good academic practice. I understand that the consequence of committing plagiarism, if proven and in the absence of mitigating circumstances may include failure in the Year or Part or removal from the membership of the University. I also certify that neither this piece of work, nor any part of it, has been submitted in connection with another assessment.

In submitting this work I can confirm that I [have/have not] used Generative AI Tools to prepare and complete my assignment.

a. I acknowledge the use of [insert AI system(s) and link] to generate materials to support my research and understanding on the topic when drafting this assignment, but none of the outputs are included in the final version.

b. I acknowledge the use of [insert AI system(s) and link] to generate materials that were included within my final assessment in modified form. I have taken responsibility to ensure that where generated materials are included, these have been referenced appropriately, in accordance with the requirements of the assignment brief.

Contents

List of Tables **iii**

List of Figures **iii**

1 Development Of Computer Simulation Methods **1**

2 Monte Carlo Methods **2**

2.1 Pseudo-Random Number Generators . . . 2

2.1.1 Mersenne Twister . . . 2

2.1.2 Well Equidistributed Long-period Linear (WELL) . . . 2

2.2 Sampling Methodology . . . 2

2.2.1 Simple . . . 2

2.2.2 Importance . . . 2

2.3 Statistical Mechanics . . . 2

2.4 The Metropolis Method . . . 2

3 Question 3? Lennard-Jones ? **3**

4 Practical Application in a Numerical Simulation **4**

4.1 Code and development . . . 4

4.2 Application to a Lennard-Jones liquid . . . 4

4.2.1 Numerical Analysis . . . 4

5 Conclusions, Findings, Prospectives **5**

6 Bibliography **6**

A Appendix A - Generative AI Tools **7**

A.1 Listing from ChatGPT . . . 7

A.1.1 ChatGPT output . . . 7

A.1.2 How I modified this material . . . 7

List of Tables

1 My caption here - example . . . 8

List of Figures

1 Development Of Computer Simulation Methods

2 Monte Carlo Methods

2.1 Pseudo-Random Number Generators

Random number generation is an important part of Monte Carlo methods. There are multiple different methods to obtaining random numbers, but the method we are focusing on within this report are called **Pseudo-Random Number Generators** (PRNGs). While they do not generate "actual" random numbers, the ones that they generate are sufficient enough in variability so that they can be used in scenarios where random numbers are needed.

The way that they achieve this is through the use of mathematical formulae, varying in levels of complexity. The algorithms can get increasingly complex, yet the simplest can consist of only a few lines of code utilising basic mathematical arithmetic. Sequences are generated which attempt to approximate random numbers, based on an initial starting *seed*.

While humans can not predict the next number in the sequences that are generated, this is not as computationally secure. This is because if the initial *seed state* is known, an identical sequence can be reproduced. Therefore, the sequences are deterministic and not as unpredictable as using systems with true chaos built in (such as radioactive atom decay, or, famously, *Cloudflare*'s wall of lava lamps).

2.1.1 Mersenne Twister

2.1.2 Well Equidistributed Long-period Linear (WELL)

2.2 Sampling Methodology

2.2.1 Simple

2.2.2 Importance

$I = \int_a^b dx f(x)$

2.3 Statistical Mechanics

2.4 The Metropolis Method

3 Question 3? Lennard-Jones ?

4 Practical Application in a Numerical Simulation

4.1 Code and development

4.2 Application to a Lennard-Jones liquid

4.2.1 Numerical Analysis

5 Conclusions, Findings, Prospectives

6 Bibliography

Frenkel, D. & Smit, B. (1996), *Understanding Molecular Simulation: From Algorithms to Applications*, Academic Press.

Gentle, J. E. (2003), *Random Number Generation and Monte Carlo Methods*, second edn, Springer.

Kopka, H. & Daly, P. W. (1999), *A Guide to LaTeX*, third edn, Pearson Education.

Sharma, V. (2023), 'Estimating pi using monte carlo methods', `https://medium.com/the-modern-scientist/estimating-pi-using-monte-carlo-methods-dbdf26c888d6`.

Wang, C. (2018), Applications of Monte Carlo Methods in Studying Polymer Dynamics, PhD thesis, University of Reading.

A Appendix A - Generative AI Tools

If you use Generative AI tools to help with your project you need to take three steps to properly acknowledge your use of these tools.

Step 1. Clearly state in their Declaration whether or not they used GAIT in their project. The Declaration should appear in the front-matter of your project, after the title page.

Step 2. In the main text you should cite your use of GAIT at appropriate points. You can reference this in the format of a personal communication or letter, including the accessed date

Step 3. Provide an AI Appendix explaining more detail

All students who have used AI should include the following in your AI Appendix:

- A description of the AI tool used (including a link),
- How the information was generated, including the prompts you used,
- The date accessed,
- Where the material is used in your project,
- How you evaluated the output from the AI tool (to check its correctness).

I suggest you include this information in the form of a table (see example in Table 1). Note that I have made this in landscape to give more space for text.

In addition, if you chose option b in your declaration (i.e. you generated materials that were included within your final assessment in modified form) you MUST include the original material generated by GAIT and briefly explain how you modified it. You can include this material in appendix subsections that you cross reference in your table (see example).

A.1 Listing from ChatGPT

A.1.1 ChatGPT output

"Here's the simplest, most common way to plot a line graph in Python using matplotlib...."

A.1.2 How I modified this material

I used my own data. I modified the axis labels and title to be specific to my data.

AI tool	Prompts used	Date accessed	Location of material in project	Evaluation of AI output	Listing of material
chatgpt.com	"Tell me how to plot a line graph in python"	12/12/2025	Approach used for plotting Figure 3	I coded it myself in python and checked the output by sketching some of the points on the plot by hand.	See A.1

Table 1: My caption here - example